
Implementation and Cross-Region Evaluation of BLISSNet on ERA5 Surface Temperature Data

Daniyal Hussain Shah

<https://www.github.com/syeddanialg/blissnet-era5-inference>

Abstract

BLISSNet is a DeepONet-style neural operator proposed for fast field reconstruction from sparse sensor measurements, originally validated on synthetic two-dimensional Navier-Stokes and quasi-geostrophic flow simulations on a fixed unit square domain. This paper presents an independent PyTorch implementation of BLISSNet and applies it to a setting the original paper does not test: real geospatial data, specifically ERA5 2-metre temperature (t2m) reanalysis over South Asia and two of its sub-regions, Punjab and Karnataka. Three models are trained independently, one per region, each with its own two-stage pipeline consisting of a SIREN trunk and Attention U-Net branch in Stage 1, followed by a frozen-trunk transformer encoder with fixed-size cross-attention in Stage 2. A full 3 by 3 cross-region evaluation is then run, testing every trained model against every region’s held-out data at 4 times super-resolution. In-domain performance is strong, with R-squared between 70.1 percent and 95.8 percent and RMSE between 0.93 K and 2.33 K across the three regions. Cross-region performance collapses entirely: R-squared falls to 0.0 percent in all six off-diagonal combinations, with RMSE and bias reaching 20 to 36 K. We explain this collapse by distinguishing between two different notions of generalization. The original paper’s zero-shot claim concerns resolution, meaning a model generalizes across grid resolutions within one fixed coordinate domain. It does not concern, and does not test, generalization across different spatial domains. In this implementation, each region-specific trunk is fit to that region’s own coordinate bounds, so evaluating a model outside its own domain is coordinate extrapolation for the SIREN trunk, compounded by a branch network that also produces out-of-distribution coefficients for unfamiliar spatial patterns. The result is a practical implication for anyone considering this architecture for geospatial use: BLISSNet is well suited to fast, repeated reconstruction over one fixed spatial domain, including zero-shot super-resolution within that domain, but is not suited, without a location-conditioned trunk or similar architectural change, to a single model deployed across multiple disjoint regions it was not trained on.

1 Introduction

Reconstructing a complete spatial field from a small set of sparse, irregularly placed observations is a common problem across climate science, remote sensing, and fluid dynamics. Classical approaches such as kriging or Gaussian process interpolation handle this task but do not scale well to large fields or to varying observation counts, and typically must be refit for every new set of sensor locations. Neural operator methods offer an alternative: rather than learning a mapping between fixed-size vectors, a neural operator learns a mapping between function spaces, so a single trained model can accept observations at different resolutions or counts, and can query the reconstructed field at arbitrary coordinates.

BLISSNet [Veremchuk et al., 2026] is a recent operator learning architecture proposed for exactly this task. It factorizes reconstruction into a trunk network, which produces a set of basis functions over the coordinate domain, and a branch network, which produces coefficients from the observed data. The reconstructed field is the weighted sum of the trunk’s basis functions, with weights given by the branch’s coefficients. The architecture is trained in two stages: Stage 1 fits the trunk and a branch that reads a fully observed grid, and Stage 2 freezes the trunk and instead trains a separate branch that reads sparse, variable-count sensor observations. The original paper validates this design on synthetic two-dimensional Navier-Stokes and quasi-geostrophic flow simulations on a fixed unit square domain, and reports zero-shot generalization across grid resolutions.

This paper re-implements BLISSNet independently in PyTorch and applies it to a real geospatial dataset: ERA5 [Hersbach et al., 2020] 2-metre temperature reanalysis over South Asia and its Punjab and Karnataka sub-regions. Our aim is to test a question the original paper does not address. The original zero-shot claim is about resolution generalization within one fixed spatial domain. We ask whether a model trained on one spatial domain can also generalize across different spatial domains, which is a natural question for anyone considering this architecture for a real-world deployment covering more than one region.

Our contributions are as follows. First, we provide a complete, independent implementation of the BLISSNet architecture and its four-term Stage 2 loss, applied to a geospatial rather than synthetic-flow setting. Second, we build a full ERA5 data pipeline, from raw GRIB files through resampling and regional slicing to the training data used here. Third, we train three region-specific models under identical hyperparameters and run a full 3 by 3 cross-region inference evaluation at 4 times super-resolution, reporting RMSE, MAE, bias, and R-squared for all nine model-region combinations. Fourth, we show that in-domain accuracy is strong and consistent with the original paper’s resolution zero-shot behavior, while cross-region accuracy collapses entirely, and we give a mechanistic explanation for why this collapse occurs and is not a moderate degradation.

2 Background: the BLISSNet architecture

BLISSNet reconstructs a scalar field $u(x)$ over a coordinate domain Ω from a set of observations. The reconstruction is written as a sum of K learned basis functions weighted by coefficients,

$$\hat{u}(x) = \sum_{k=1}^K c_k \phi_k(x), \quad (1)$$

where $\phi_k(x)$ are basis functions produced by a trunk network as a function of coordinates $x \in \Omega$, and c_k are coefficients produced by a branch network from the observed data. This factorization is the defining structure of a DeepONet: the trunk carries the geometry and coordinate dependence, while the branch carries the dependence on the specific observation.

Stage 1: full-field reconstruction. In Stage 1, the branch is given a fully observed grid and must produce K coefficients from it. The branch consists of an Attention U-Net encoder, which down-samples the input grid through convolutional and residual blocks with multi-head self-attention at the bottleneck, followed by a transformer and linear decoder that maps the resulting embedding to a vector of K coefficients. The trunk is a coordinate-MLP with sinusoidal activations, which takes a coordinate mesh grid as input and outputs K basis function values at every coordinate. Trunk and branch are trained jointly, minimizing the mean squared error between the reconstructed field and the ground-truth grid. Figure 1 shows this Stage 1 architecture.

Stage 2: sparse-observation interpolation. In Stage 2, the trunk and the Stage 1 coefficient decoder are frozen. A new branch is trained that reads a sparse, variable-count set of sensor coordinates and values instead of a full grid. This branch uses a transformer encoder to embed the sparse observations, followed by a fixed-size cross-attention block that queries a constant grid of learned positions, producing a constant-size embedding regardless of how many sensors were observed. This avoids the quadratic cost of attending directly over a variable and potentially large number of sensor points. A decoder then maps the fixed-size embedding to the

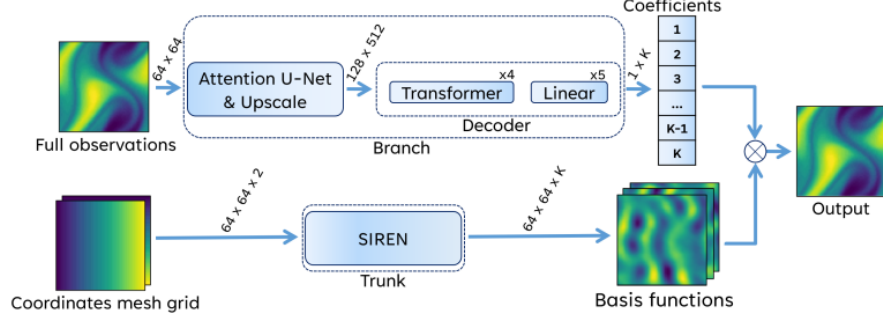


Figure 1: Stage 1 architecture. A fully observed grid is encoded by an Attention U-Net and decoded through a transformer and linear layers into K coefficients. A coordinate mesh grid is passed through a SIREN trunk to produce K basis functions. The output field is their weighted sum.

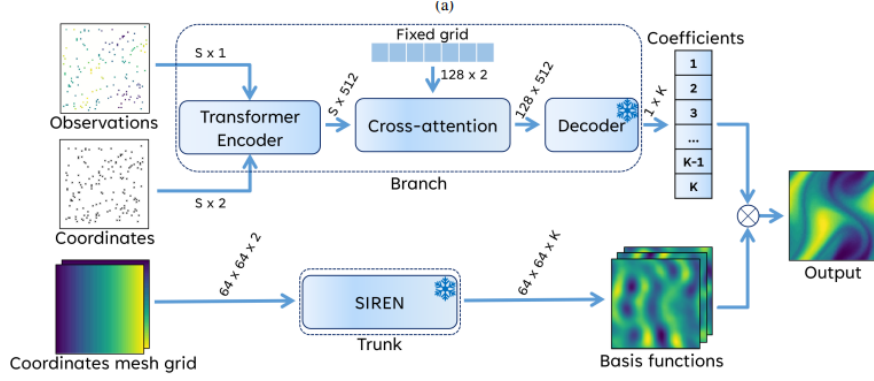


Figure 2: Stage 2 architecture. The trunk and coefficient decoder are frozen from Stage 1. A transformer encoder reads sparse, variable-count sensor coordinates and values, and a fixed cross-attention block against a constant grid of learned positions produces a constant-size coefficient vector.

same K -dimensional coefficient vector used in Stage 1, and the field is reconstructed using the same frozen trunk basis functions. Figure 2 shows this Stage 2 architecture.

Stage 2 loss. Stage 2 training uses a four-term loss,

$$\mathcal{L} = \lambda_1 \mathcal{L}_{cp} + \lambda_2 \mathcal{L}_{coef} + \lambda_3 \mathcal{L}_{emb} + \lambda_4 \mathcal{L}_{gt}, \quad (2)$$

where $\mathcal{L}_{cp}$ is the control-point loss, the mean squared error between the reconstructed field’s value at the observed sensor locations and the true sensor values; $\mathcal{L}_{coef}$ is the coefficient loss, the mean squared error between the Stage 2 coefficients and the Stage 1 coefficients on the same sample, used as a target; $\mathcal{L}_{emb}$ is the embedding loss, the mean squared error between the intermediate embeddings produced by the Stage 2 and Stage 1 branches; and $\mathcal{L}_{gt}$ is the ground-truth loss, the mean squared error between the full reconstructed field and the ground-truth grid. This combination supervises Stage 2 both at the sparse input locations and against the full Stage 1 solution, so that the sparse-observation branch is pulled toward reproducing the same coefficients, embeddings, and field values that the full-grid branch would have produced.

A property of this design, central to the findings of this paper, is that the trunk network is tied to a single coordinate domain. Because the SIREN trunk is a continuous function of coordinates rather than a fixed-resolution lookup table, a trunk fit to one coordinate domain can be queried at resolutions finer or coarser than anything seen during training, which is the resolution zero-shot behavior the original paper reports. This says nothing, however, about querying the trunk at coordinates outside the domain it was fit to; that is an extrapolation question, not a resolution question, and is exactly what this paper tests.

Table 1: Regional datasets used in this work. All three regions share 1460 six-hourly timesteps over the year 2025 and differ only in spatial bounds and grid resolution.

Region	Bounds (lat, lon)	Grid ($H \times W$)	Points ($T \times H \times W$)
South Asia	5 to 38°N, 60 to 98°E	133×153	$\approx 29.7\text{M}$
Punjab	27.5 to 34°N, 69 to 75.5°E	27×27	$\approx 1.06\text{M}$
Karnataka	11.5 to 18.5°N, 74 to 78.5°E	29×19	$\approx 0.80\text{M}$

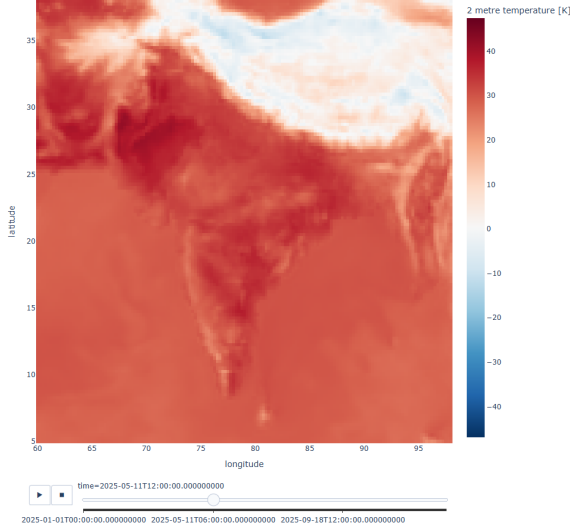


Figure 3: A single 6-hourly snapshot of the South Asia t2m field used for training and evaluation, spanning 5 to 38°N and 60 to 98°E. Color indicates 2-metre temperature in Kelvin, converted to Celsius for display.

3 Dataset

We use ERA5 [Hersbach et al., 2020] 2-metre temperature (t2m) reanalysis data, produced by the Copernicus Climate Change Service and obtained through the Copernicus Climate Data Store via cfgrib. The raw data covers the full year 2025 at hourly resolution over a South Asia bounding box of 5 to 38 degrees north latitude and 60 to 98 degrees east longitude.

The data preparation pipeline proceeds as follows. Raw hourly GRIB files are loaded and unused coordinate variables, namely surface, step, valid time, and number, are dropped. The hourly series is resampled to 6-hour means, reducing 8760 hourly timesteps to 1460 six-hourly frames. Two sub-regions, Punjab and Karnataka, are then sliced from the South Asia grid by latitude and longitude bounds. All three resulting datasets are exported to NetCDF for use in training and inference. Table 1 summarizes the spatial extent and grid size of each region; all three regions share the same 1460 timesteps and differ only in spatial extent and resolution.

Figure 3 shows one snapshot of the South Asia t2m field. The field shows the expected seasonal and geographic structure of surface temperature over South Asia, including cooler high-elevation regions along the northern Himalayan boundary and warmer conditions across the central and southern plains.

For every training sample used in Stage 2, between 0.5 percent and 5 percent of grid points are subsampled at random to form the observed sparse sensor set, so that the Stage 2 encoder is forced to interpolate from a variable and genuinely sparse input rather than a near-complete grid.

Table 2: Training hyperparameters, held identical across all three region-specific models.

Parameter	Value	Parameter	Value
Embedding dimension	512	Basis functions K	512
Attention heads	8	Batch size	32
Learning rate	1×10^{-5}	Dropout	0.1
SIREN hidden size / layers	512 / 4	SIREN ω_0	30
U-Net base channels	64	Epochs per stage	15
Train / test split	90 / 10	$(\lambda_{cp}, \lambda_{coef}, \lambda_{emb}, \lambda_{gt})$	100, 40, 0.01, 0.5

4 Experimental setup

Models. Three BLISSNet models are trained independently, one per region: South Asia, Punjab, and Karnataka. Each model has its own trunk, branch, and Stage 2 encoder, trained from scratch on that region’s data only. Coordinates for every model are normalized globally to the range $[-1, 1]$ before training, so that all three regions’ coordinate bounds lie within the same normalized range. This choice was made specifically to give cross-region generalization the best possible chance, since it removes any trivial mismatch in coordinate scale between regions; as discussed in Section 5, it does not resolve the underlying failure.

Hyperparameters. All three models are trained with identical hyperparameters, listed in Table 2, so that region is the only variable that differs across the three training runs. Each model is trained for 15 epochs per stage.

Evaluation protocol. After training, each of the three trained models is run at 4 times super-resolution against the held-out test data of all three regions, giving 9 model-region combinations in total. Three of these are in-domain, meaning the model is evaluated on its own training region, and six are cross-region, meaning the model is evaluated on a region it never saw during training. For every combination we report root mean squared error (RMSE), mean absolute error (MAE), bias, and the coefficient of determination (R-squared), all computed on held-out test frames in the original Kelvin scale.

5 Results

5.1 Training behavior

Figure 4 shows training curves for the South Asia model. Stage 1 loss falls rapidly within the first two to three epochs and then flattens, with train and test RMSE converging to roughly 2.2 to 2.8 K and R-squared reaching approximately 95 to 96 percent by the final epoch. Stage 2 loss also decreases steadily across all four loss components, control-point, coefficient, embedding, and ground-truth, with the Stage 2 field-level R-squared reaching approximately 93 to 96 percent. Figure 5 shows the corresponding Stage 1 loss curves for the Punjab and Karnataka models, and Stage 2 curves for Punjab; all three regions show the same overall pattern of rapid early convergence followed by a long, stable plateau, indicating that training was stable and converged within the 15 epochs used for every stage and every region.

5.2 Cross-region inference

Table 3 reports the full 3 by 3 cross-region evaluation. The diagonal entries, where a model is evaluated on its own training region, show strong and consistent accuracy: R-squared ranges from 70.1 percent (Karnataka model on Karnataka) to 95.8 percent (South Asia model on South Asia), with RMSE between 0.93 K and 2.33 K. This in-domain behavior is consistent with the resolution zero-shot generalization reported in the original BLISSNet paper, since every diagonal entry is itself a 4 times super-resolution result relative to the training grid.

The off-diagonal entries, where a model is evaluated on a region it was not trained on, show a complete collapse rather than a moderate drop in accuracy. R-squared falls to exactly 0.0 percent

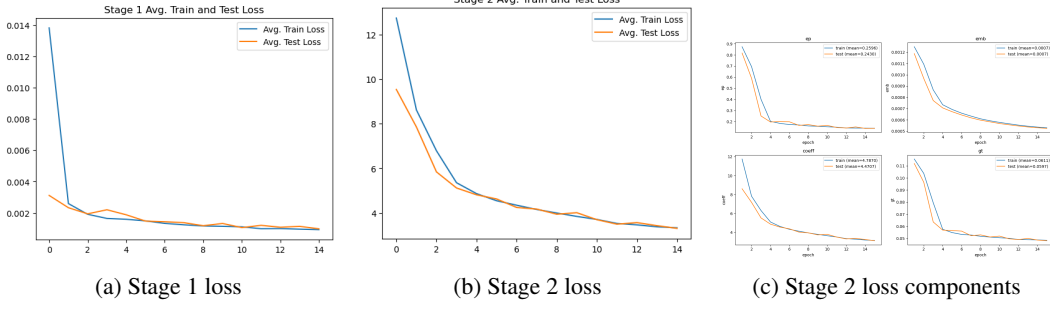


Figure 4: South Asia model training curves. (a) Stage 1 average train and test loss across epochs. (b) Stage 2 average train and test loss. (c) The four individual Stage 2 loss components from Equation 2: control-point (ep), embedding (emb), coefficient (coeff), and ground-truth (gt).

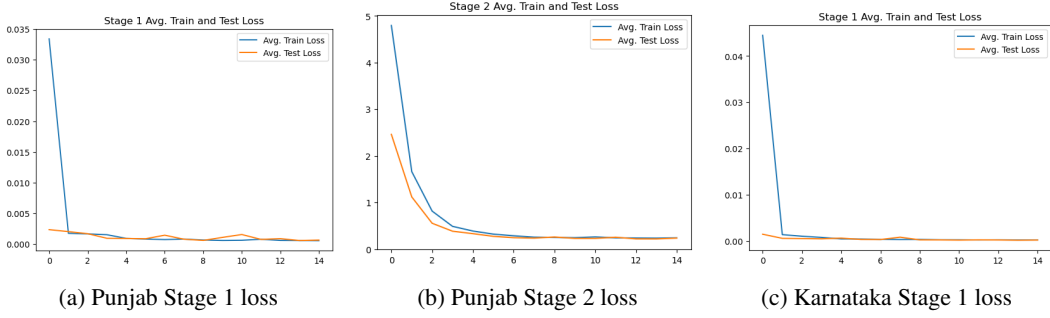


Figure 5: Training loss curves for the Punjab and Karnataka models. All three region-specific models were trained under identical hyperparameters (Table 2) and show the same overall convergence pattern as the South Asia model in Figure 4.

in all six cross-region combinations. RMSE and bias both grow by roughly an order of magnitude relative to the corresponding diagonal entry, reaching between 8.1 K and 35.9 K RMSE, with the Punjab-trained model evaluated on Karnataka showing the largest error at 35.9 K RMSE and a bias of -35.8 K. Figure 6 visualizes this pattern directly: R-squared is high only on the diagonal and at or near zero everywhere else, and RMSE is low only on the diagonal and substantially elevated everywhere else, for every one of the three models.

5.3 Why cross-region generalization fails

The original paper’s zero-shot claim concerns grid resolution: a model trained on a fixed coordinate domain, such as the unit square used in the synthetic flow experiments, reconstructs accurately at grid resolutions it was never explicitly trained on. This is possible because the SIREN trunk is a continuous function of coordinates rather than a fixed-resolution grid, so querying it more densely within the same domain is interpolation in coordinate space, not extrapolation. The original paper does not claim, and does not test, generalization to a different spatial domain than the one the trunk was fit to.

In this implementation, each of the three region-specific models has its own independently fitted trunk network, tied to that region’s coordinate bounds, despite all three models sharing the same global $[-1, 1]$ coordinate normalization. Evaluating a model outside its trunk’s fitted coordinate domain is therefore coordinate extrapolation for the SIREN trunk, a known weak point of coordinate-MLP architectures, since the trunk has only ever seen coordinate values corresponding to its own region’s geography and has no basis for producing meaningful basis function values elsewhere in the shared normalized range. This failure is compounded by the branch network, which has no explicit notion of geographic location and was trained only on spatial patterns drawn from its own region; when given a spatial input pattern from an unfamiliar region, the branch also produces out-of-distribution coefficients. Because both halves of the trunk-branch product are simultaneously

Table 3: Cross-region evaluation. Each row is a model trained on one region; each column group is the region it is evaluated on, at 4 times super-resolution on held-out test frames. Diagonal entries (model evaluated on its own training region) are shown in bold. RMSE, MAE, and bias are in Kelvin; R-squared is in percent.

Trained on	South Asia (test)				Punjab (test)				Karnataka (test)			
	RMSE	MAE	R ²	Bias	RMSE	MAE	R ²	Bias	RMSE	MAE	R ²	Bias
South Asia model	2.33	1.65	95.8	-0.24	8.13	7.16	2.4	5.13	3.27	2.64	0.0	1.52
Punjab model	25.52	22.20	0.0	-21.32	1.78	1.35	82.3	0.63	35.94	35.78	0.0	-35.78
Karnataka model	19.10	16.72	0.0	-11.63	19.92	18.35	0.0	-17.42	0.93	0.75	70.1	0.20

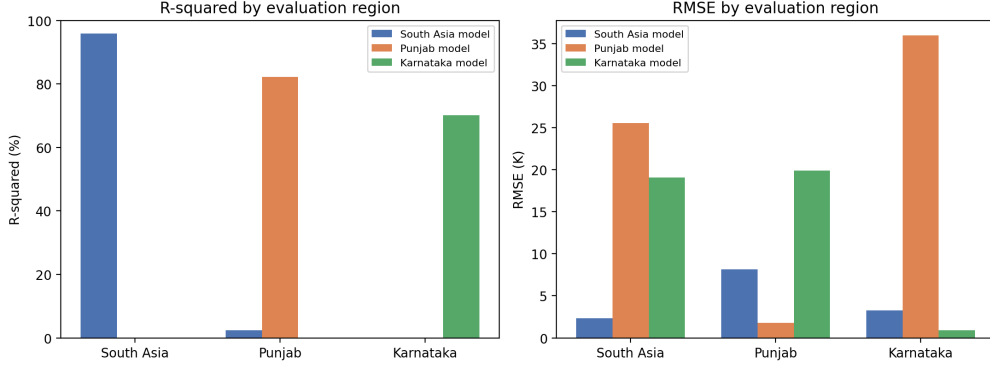


Figure 6: Cross-region R-squared and RMSE for each of the three trained models against each of the three evaluation regions, corresponding to the rows and column groups of Table 3. R-squared is high only when the evaluation region matches the training region; every off-diagonal combination collapses to 0 percent R-squared, and RMSE rises correspondingly.

unreliable outside the training region, the result is a sharp collapse rather than the graceful degradation that a partial mismatch, for example a resolution change alone, would be expected to produce.

We note that using a shared global coordinate normalization across all three models, rather than a per-region normalization, was a deliberate design choice intended to give cross-region generalization its best possible chance, since it at least aligns the coordinate scale and sign conventions across regions. This did not resolve the failure. The result indicates that the trunk is not simply failing due to a coordinate scale mismatch, but has instead learned a representation that is tied to the specific geospatial structure, meaning the particular arrangement of temperature patterns relative to coordinates, of the single region it was trained on, rather than a coordinate-general basis that would transfer to an unrelated region occupying the same normalized coordinate range.

6 Discussion and practical implications

The results in this paper support a specific and practically useful conclusion about the scope of BLISSNet in this implementation. Within a fixed spatial domain, the architecture performs well: R-squared of 70 to 96 percent and RMSE below 2.4 K across three different regions of varying size and climate, at 4 times super-resolution relative to the training grid, is a strong result for a sparse-observation reconstruction task on real reanalysis data. This makes the architecture well suited to applications that require fast, repeated reconstruction over one fixed spatial domain from sparse or variable-count observations, including tasks that benefit from zero-shot super-resolution within that domain, for example filling in a higher-resolution temperature field over a single country or weather station network from a modest number of sensors.

The architecture, as implemented here and following the original paper’s design, is not suited to a single model deployed across multiple disjoint regions it was not trained on. A user who trains one BLISSNet model per region, as we did, cannot expect that model to transfer to a new region without retraining; the collapse in Section 5 is total rather than partial, and is not mitigated by a shared global

coordinate normalization. Extending BLISSNet to genuinely support multiple regions with a single model would likely require an architectural change such as a location-conditioned trunk, for example a trunk that receives an explicit region or location embedding in addition to local coordinates, so that the basis functions it produces can depend on which region’s geography those coordinates belong to. We leave this extension to future work.

Limitations. This study covers one geophysical variable, 2-metre temperature, over one calendar year and three regions all located in South Asia, so the regions compared here share broadly similar climate and are not maximally different in the way, for example, comparing a tropical region to a polar region would be. It is possible that a more extreme domain shift would produce an even sharper collapse, or conceivably a different failure pattern; we did not test this. Each model was trained for 15 epochs per stage, which was sufficient for the loss curves in Section 5 to plateau, but longer training or a larger basis dimension K were not explored and might shift the in-domain results somewhat, although we would not expect this to change the qualitative cross-region finding, since the failure mode we identify is structural, tied to what the trunk and branch can represent outside their training domain, rather than a symptom of undertraining.

Acknowledgments

This implementation follows the architecture and training methodology described in the original BLISSNet paper [Veremchuk et al., 2026]. All code in this work, including the PyTorch implementation, the ERA5 data pipeline, the training runs, and the cross-region inference experiments, was independently written and run by the author. The original paper’s authors are not affiliated with this work, and the results reported here, including the cross-region generalization findings, have not been reviewed or endorsed by them.

This work uses ERA5 reanalysis data, produced by the Copernicus Climate Change Service (C3S) and obtained from the Copernicus Climate Data Store (CDS). Generated using Copernicus Climate Change Service information (2025). Contains modified Copernicus Climate Change Service information (2025). Neither the European Commission nor ECMWF is responsible for any use that may be made of the Copernicus information or data it contains. ERA5 data is distributed under the Copernicus License (CC-BY), which permits free reuse, redistribution, and modification, subject to the attribution requirements above.

References

- Veremchuk, M., Scott, K. A., and Pan, Z. BLISSNet: Deep Operator Learning for Fast and Accurate Flow Reconstruction from Sparse Sensor Measurements. *arXiv preprint arXiv:2602.24228*, 2026.
- Hersbach, H., Bell, B., Berrisford, P., et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020.